Methodology Pipeline: Domain-Ablation Study of Student Stress Level

Dataset Collection
StressLevelDataset.csv: 1,100 students
20 predictors + 1 target (stress_level: 0 / 1 / 2)

Data Preprocessing & Audit
Missing values = 0, duplicate rows = 0
Class-count check; shape (1100, 21) verified

Domain Mapping
20 features -> 5 domains (4 features each):
Psychological | Physiological | Environmental |
Academic | Social

Is data balanced?
No
Balancing
Skipped (SMOTE / resampling):
classes already balanced
(373 / 358 / 369)
Yes

Train-Test Split
Stratified 80/20 split (seed = 42)
Train = 880 | Holdout = 220

Fit on Training Data
StandardScaler (z-score)
fitted on training folds only
(Pipeline: scaler -> classifier)

Transform Test Data
Fitted scaler applied to the
validation fold / 220-sample
holdout set (no leakage)

Model Evaluation & Optimization
Candidates: Logistic Regression, Random Forest, Gradient Boosting
RepeatedStratifiedKFold (5 folds x 3 repeats)
Hyper-parameter tuning: GridSearchCV (nested, inner 3-fold)

Performance Measurement
Weighted F1, Macro F1, Balanced Accuracy, MCC
Mean +/- SD over 15 folds -> best model selected (weighted F1)

Domain Ablation
11 conditions: All features, Drop-one (x5), Only-one (x5)
Same CV folds; tuning repeated inside every fold

Statistical Validation
Wilcoxon signed-rank (All vs Drop-domain), bootstrap 95% CI
Holm correction; multi-seed robustness (5 seeds, Spearman)

Final Model & Holdout Test
Tuned model refit on full training set
Evaluated on the 220 holdout samples (F1, Bal. Acc., MCC)

XAI Analysis
SHAP: global, class-wise and domain-level importance
Spearman: SHAP domain ranking vs ablation ranking

What-if Analysis
+/-1 step change on 6 modifiable features
(sleep, social support, study load, noise, peer pressure, bullying)
-> change in predicted P(high stress)

Results & Reporting
Tables (CSV) and figures (PNG) exported for the paper